4A

L	R/unit length	C/unit length	Intrinsic date delay	
500	0.08	0.2	35	
2000	0.08	0.2	35	
5000	0.08	0.2	35	
micrometer	ohms/micrometer	fF/micrometer	ps	

Wire delay = 1/2*RLCL

L	RL	CL	Wire delay	
500	40	100	2000	2ps
2000	160	400	32000	32ps
5000	400	1000	200000	200ps
			fs	

I apologise for the trouble! Due to my disability, my wrists and hands are not working well at the moment so I must type all my work. I hope that I have formatted my work in a clear enough manner to mark, but if not, please let me know and I will explain my reasonings over a Zoom call or in person.



4B

gate delay = 35ps

35ps =	0.5*RCL^2	
35ps*2/RC=	L^2	
4375000	equals L^2	(calculating with femtoseconds)
2091.650066	**Equals L**	**in micrometers**

4C

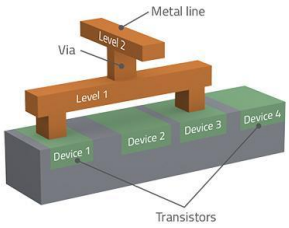
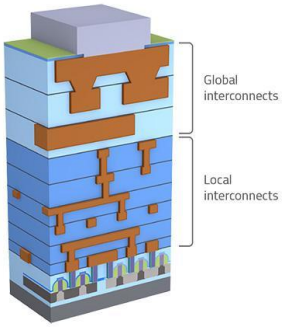
L	R/unit length	C/unit length
500	0.08	0.2
2000	0.08	0.2
5000	0.08	0.2
micrometer	ohms/micrometer	fF/micrometer

L	C/unit length	CL
500	0.2	100
2000	0.2	400
5000	0.2	1000
micrometer	fF/micrometer	fF

Pdyn = activity factor *C/L*L*Vdd^2*f

L	Cap. per unit length	Vdd	Activity factor	Freq	**Pdyn**
500	0.2	0.9	0.1	1	**8.1**
2000	0.2	0.9	0.1	1	**32.4**
5000	0.2	0.9	0.1	1	**81**
micrometer	fF/micrometer	V		GHz	microwatts

Image from: https://newsroom.lamresearch.com/Tech-Brief-An-Introduction-to-Interconnects

4D	 	
	Local Interconnect refers to thin, closely packed wire connections that connect close-proximity transistors. Global Interconnect refers to the thicker, longer wires that connect different areas of the chip and are further apart. As technology improves (ie as the transistors are able to be more closely packed together and reduce in size) the length and width of local interconnect wires also reduce. However, because global interconnect wires remain exceptionally long, spanning the chip even, their length does not decrease in the same way.	
	Information from:	https://newsroom.lamresearch.com/Tech-Brief-An-Introduction-to-Interconnects
4E	Repeater insertion would increase propagation delay, along with costs and area usage.	
	Using a wider wire would increase costs, area consumption, and the possibility of crosstalk between wires	